8	**C**	Let X be the gas compressed isothermally (no change in temp). There is heat exchange between the gas and the surroundings. Let Y be the gas compressed and is isothermally isolated from surroundings. There is no heat exchange between gas and the surroundings. For X: Since temperature does not change due to heat exchange between gas and the surroundings, $p_1V = p_2(0.5V)$, $p_2 = 2\,p_1$ For Y: Using $pV = nRT$, when V decreases, pressure increases and leads to larger speed of the molecules. This leads to higher temperature of gas. A: No heat is given to both gases during compression. B: Internal energy of Y is higher due to higher temperature of gas that leads to higher average KE. C: Since mass and volumes of gases are the same, there is no change in density of gases. D: Since work done on gases depends on the pressure and the change in volume and there is no information on how pressure varies for both gases during compression, the work done on gases may not be the same.
9	**B**	The following options are wrong